HOW TO

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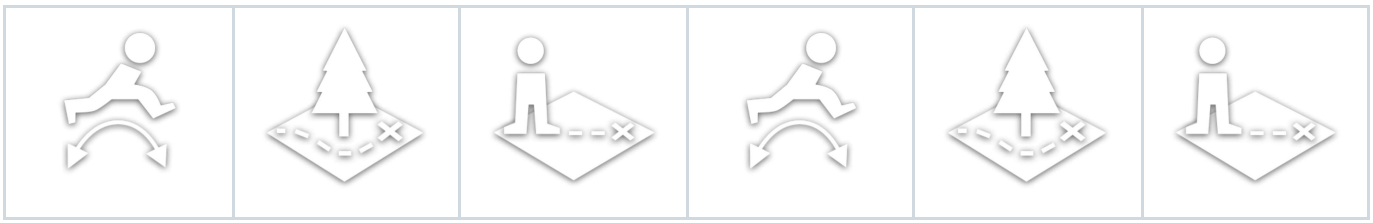
NAVMESH PLUS



NavMesh Extensions System

by h8man

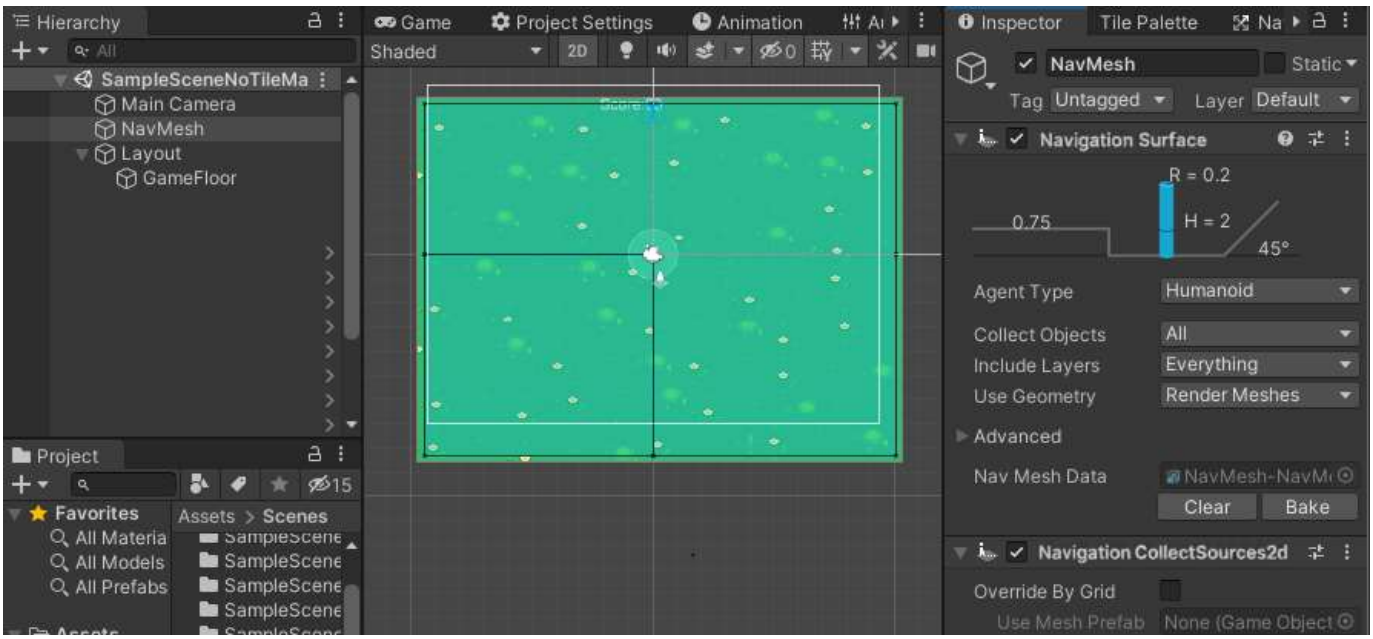
INTRO

This short tutorial will focus on achieving the most basic functionality with `NavMeshPlus - NavMesh Extensions System` for 2D games, by using `NavigationCollectSources2d` and `NavigationModifier` extension components.

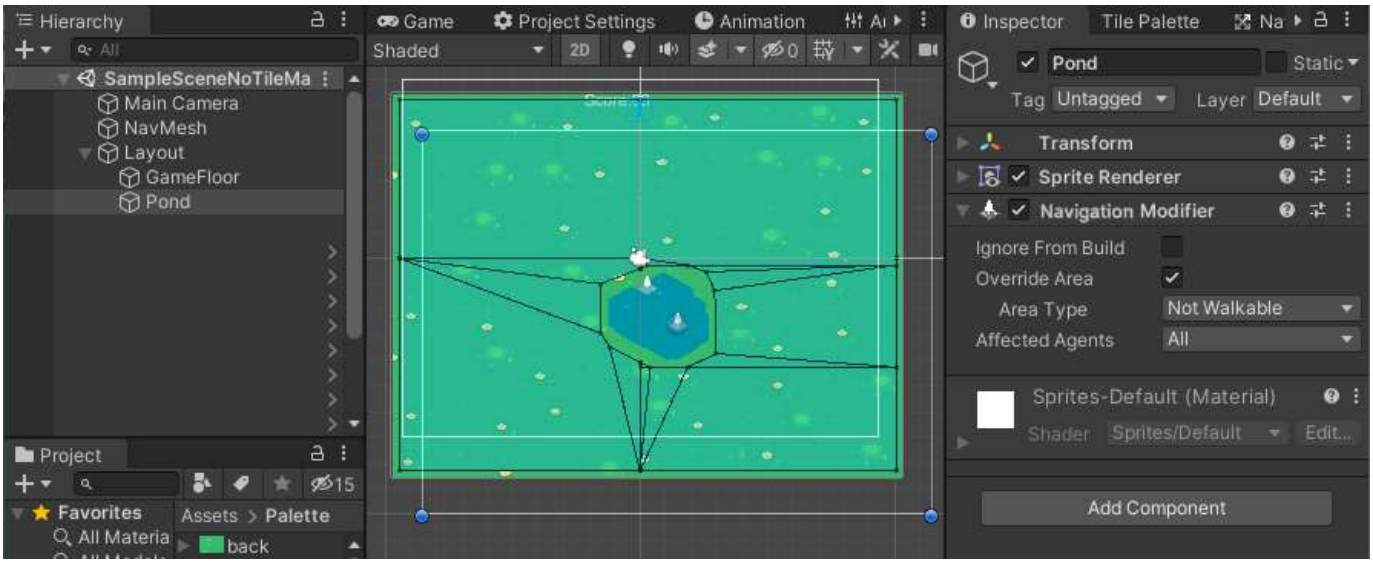
To begin with, Create a new project from Unity's "2D Template". Download NavMeshPlus and copy NavMeshComponents into your Assets folder. Now, in your scene, right click and select `Create Empty` from a menu to add an empty `GameObject` into scene's root. Name your new empty `GameObject`, for example as "NavMesh".

NAV MESH BASICS

Create a new root `GameObject` named "Layout" where we will store our level layout and add child `GameObject` as "GameFloor". Drop any sprite onto "GameFloor", `SpriteRenderer` component will be automatically added, now add `NavigationModifier` component to the object to mark it as Navigation source. With your "NavMesh" object selected, in the inspector click `Add Component` and choose `NavigationSurface`. If you hit bake nothing will happen, cause we need to add another component `NavigationCollectSources2d`, click `Rotate Surface to XY` and then `Bake` to bake walkable plain.



Now add another object to "Layout", call it "Pond", drop new smaller sprite and similarly click `Add Component`, choose a `NavigationModifier` component. In the `NavigationModifier` component check `Override Area` and in the `Area Type` field choose `Not Walkable`. Now hit `Bake` again (within the "NavMesh" object). You should see that the "Pond" have been carved out of the Navigation areas.

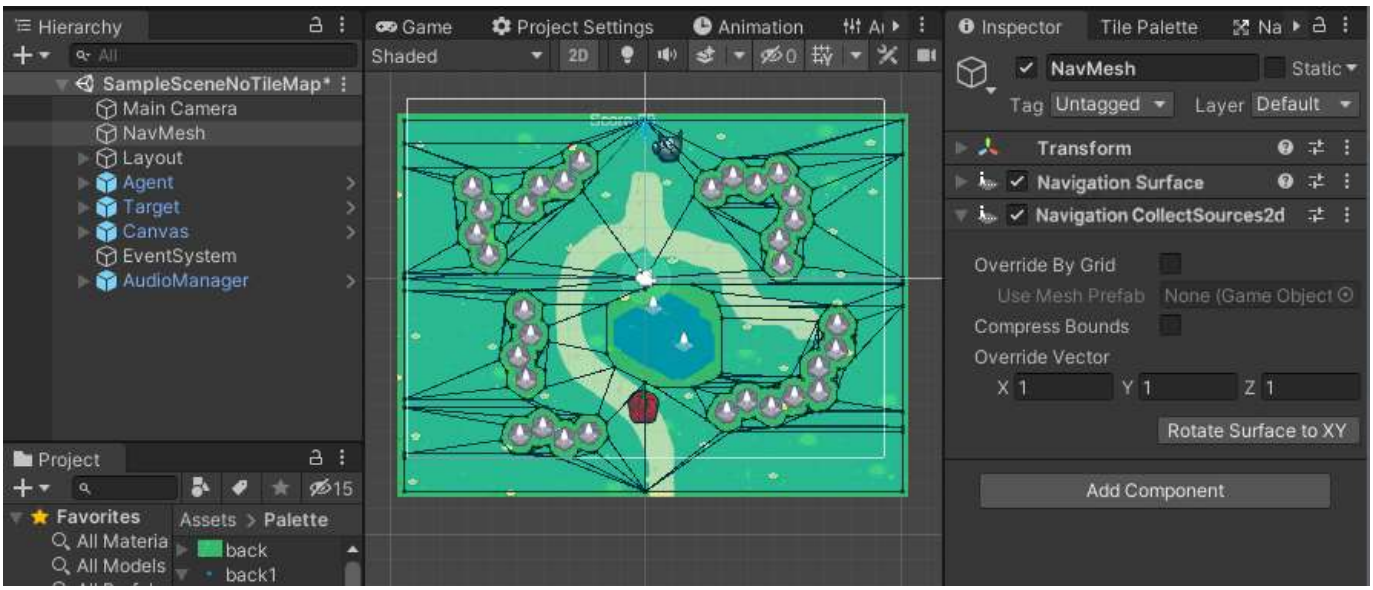


At this point, you can add GameObjects with a `NavMeshAgent` component, which are capable of interacting with (i.e., walking on) your NavMesh.

By default `NavMeshAgent` component tends to rotate the `GameObject` as soon as you hit play. This can be undesirable (as in, your object may be rotate away from your plane of view and hence become invisible), so you should fix its rotation by adding something like the following to a new or existing script:

```
void Start()    {
    var agent = GetComponent<NavMeshAgent>();
    agent.updateRotation = false;
    agent.updateUpAxis = false;
}
```

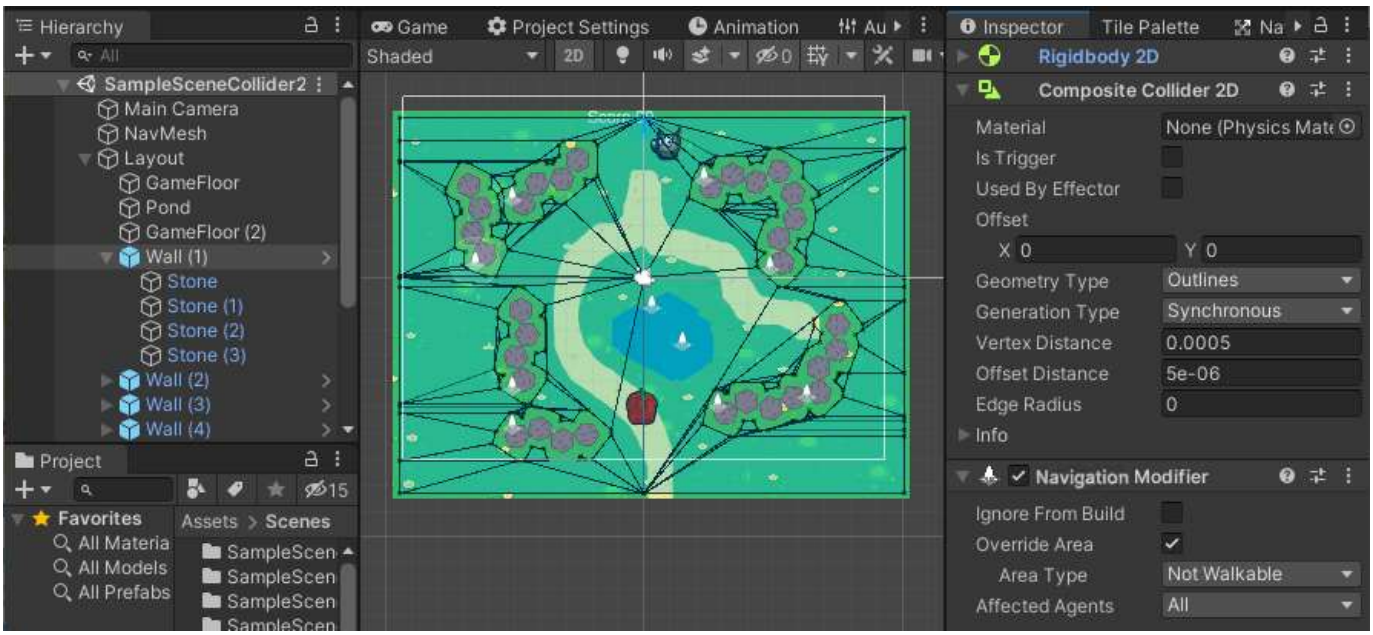
Once this is added to the `Start` method for all of your agents, you are ready for action.



Important note: *Currently there is no way to define world bound except to introduce explicit world bound* `GameObject` *with* `NavigationModifier`

GEOMETRY OPTIONS

With new API introduced in Unity 2019.3 there is an ability to choose which geometry source to use. Place all 2D colliders with `NavigationModifier` added into "Layout" GameObject as child, change `Use Geometry` option from `Render Meshes` to `Physics Colliders` and hit bake.



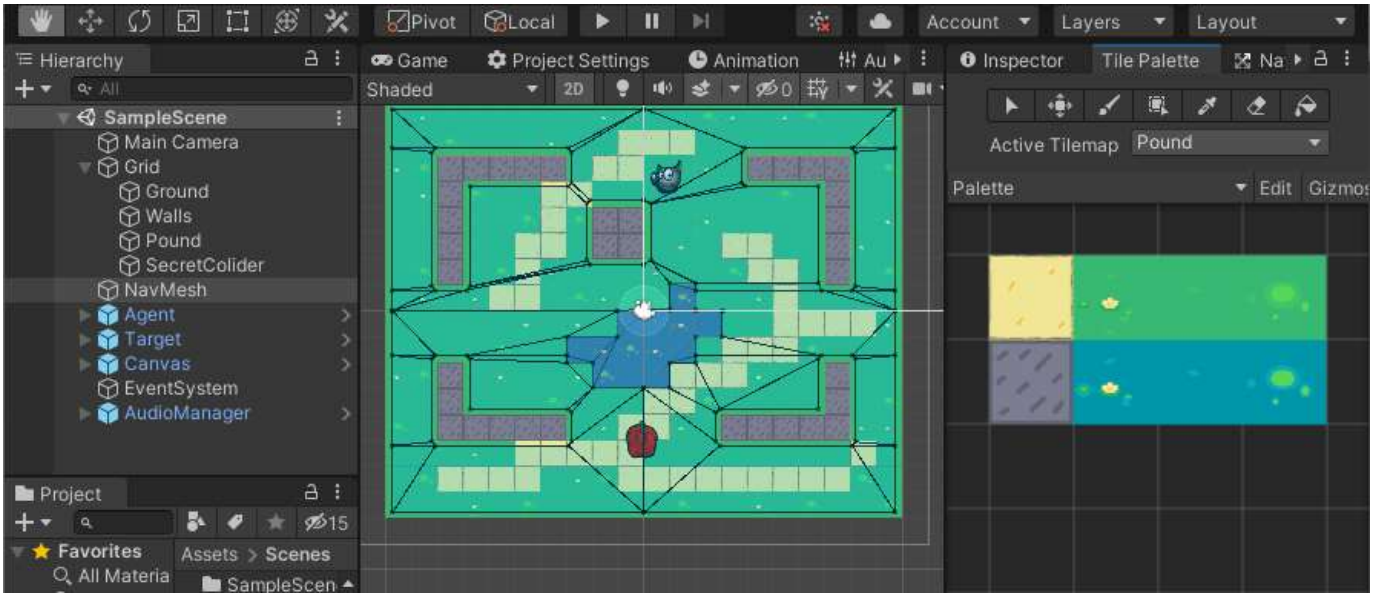
Important note: *It is discovered in the [Unity Docs]() for the CreateMesh method that the EdgeCollider2d will create a mesh if it has an edgeRadius set.*

Important note: *By default* `Use Geometry` *option is set to* `Render Meshes` *, which mean* `NavigationSurface` *use sprite's Mesh to build its navigation surface, so surface will stick to sprite "outline". The other option available is* `Physics Colliders` *, this option allows navmesh to build surface form 2D colliders polygons. Options are mutually exclusive.*

TILED

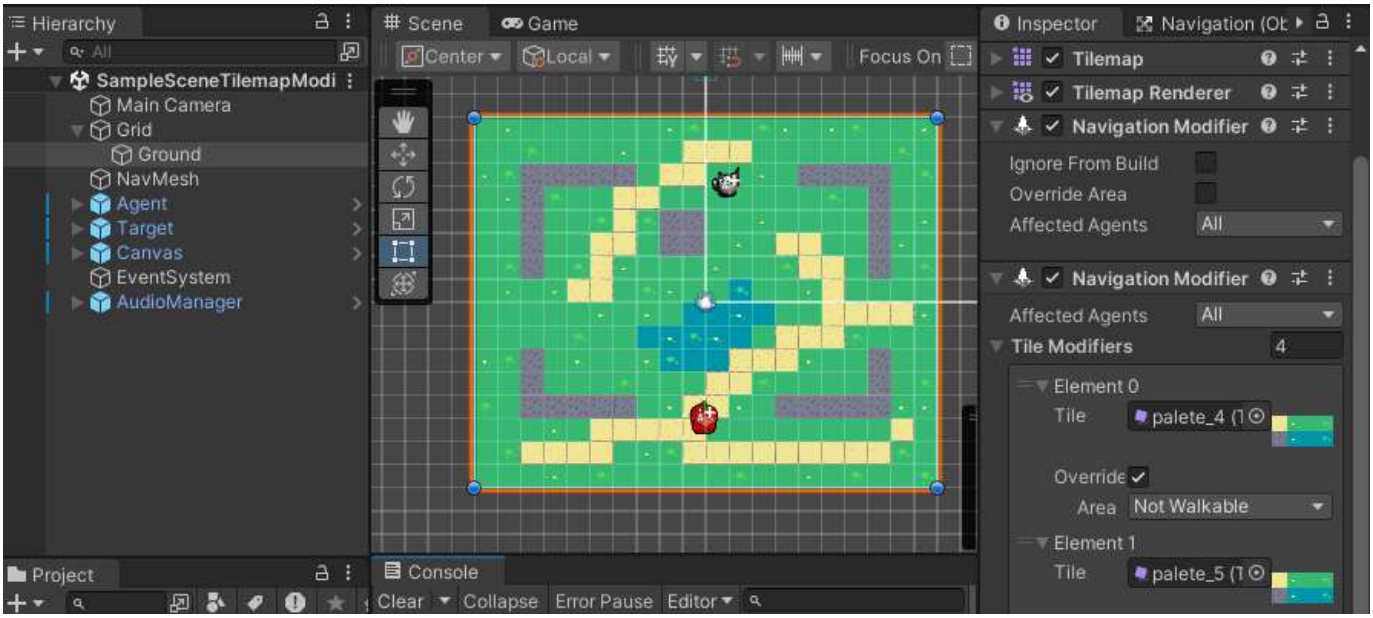
With `NavigationCollectSources2d` you can bake objects that are a part of `Tilemap` system. Add a `Tilemap` GameObject to your scene (By right-clicking your Scene hierarchy and choosing the `2D Object` folder, and then `Tilemap` within). It should create a `Grid` object in the root of your scene and child `Tilemap` game object. Add two more `Tilemap` s (for a total of 3) and and name them "Ground", "Walls" and "Pond".

Using sprites imported into your palette, choose "Ground" as the `Active Tile Map` , and paint the ground in your scene view. Change `Active Tile Map` in your palette to "Walls" to paint some obstacles. Then select "Pond" and do the same, drawing a body of water. As with basics navmesh add `NavigationModifier` to each of tilemaps, override its areas to walkable and not-walkable respectively. Bake the navmesh.



TILE MODIFIER

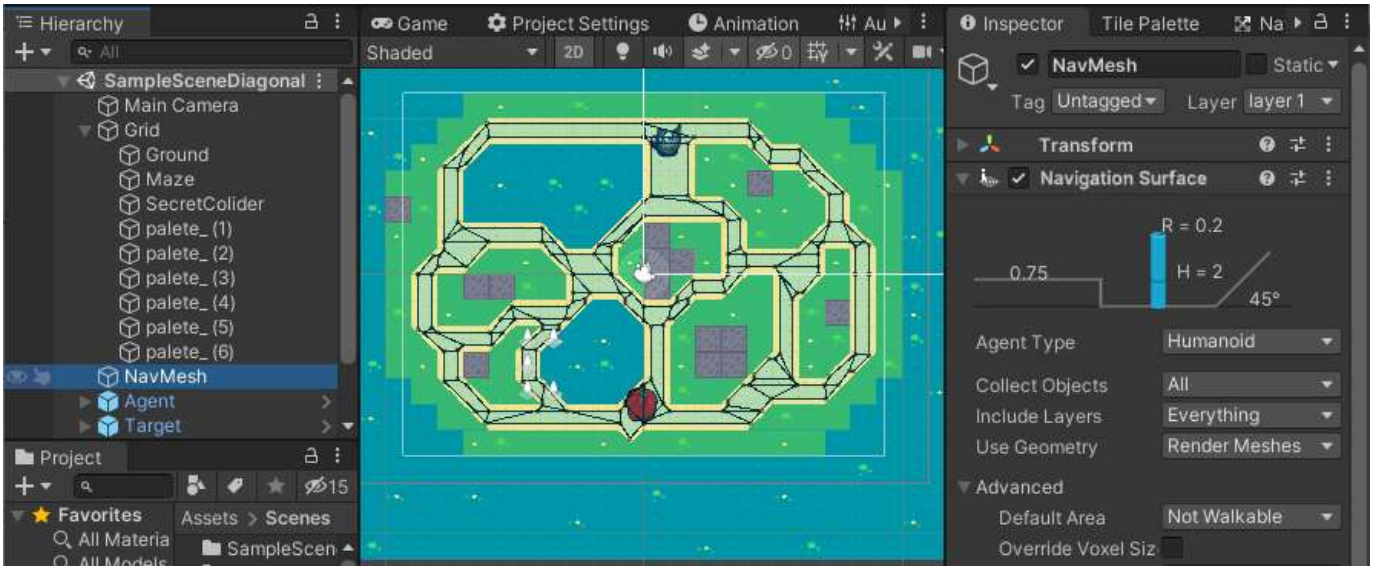
As each `Tile` represent properties of the world it is possible to assign navigation modifier to each tile in `TileSet` . To do that use `NavigationModifier` and `NavigationModifierTilemap` on asingle `Tilemap` paited form existing tilest. Add new element to the list inside `NavigationModifierTilemap` and chosse tile from assets. Now overide each tile on top of `NavigationModifier` , or leave as it is.



Hit bake and get seme result as in chapter above, exept with no layers on a single `TileMap` .

MAZES

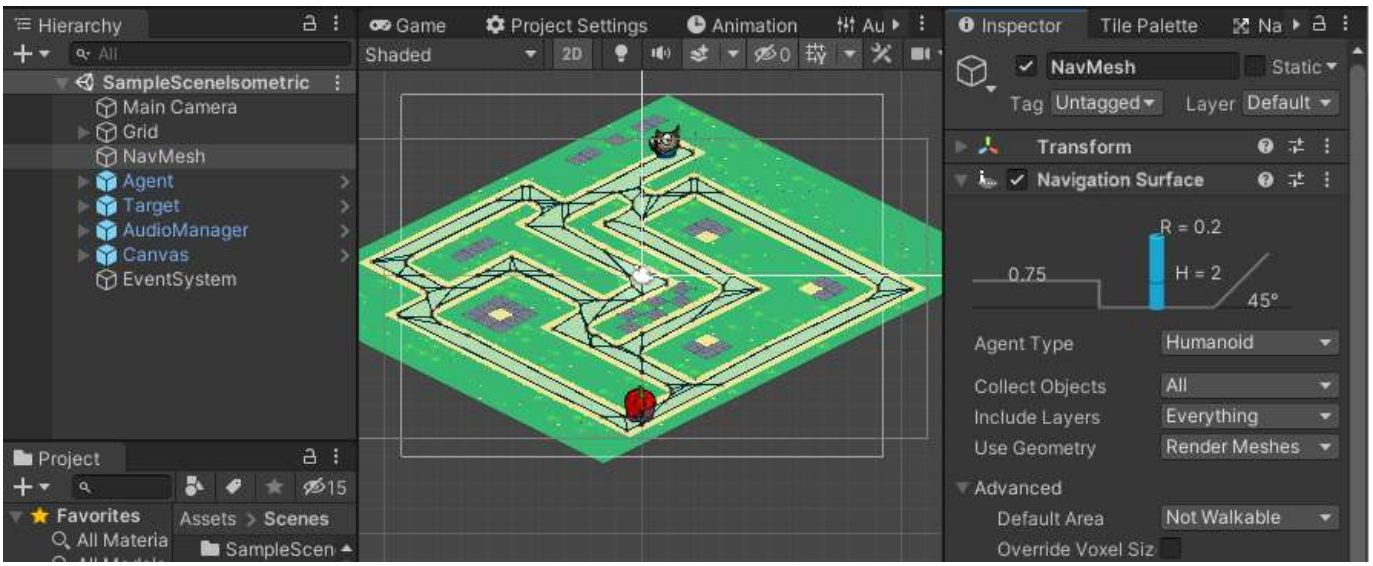
Another common task is not to carve out obstacles, but to navigate through a maze, where our agents can only follow a strict path. To do that, Bake your "NavMesh" with `Default Area` selected to `Not Walkable`, and the `NavigationModifier` component set to be `Walkable`, and you will get this:



Any sprite within the `Grid` object with a `NavigationModifier` will be added into Bake process. (Sprite must have a mesh). You can build bridges or barricades, and these objects will be added into your `NavMesh` as static data.

OTHER GRIDS

Some games want to use hexagonal, isometric, or diamond-shaped grids. NavMeshPlus supports baking other types of grid as well, as long as spites share the shape as your grid.

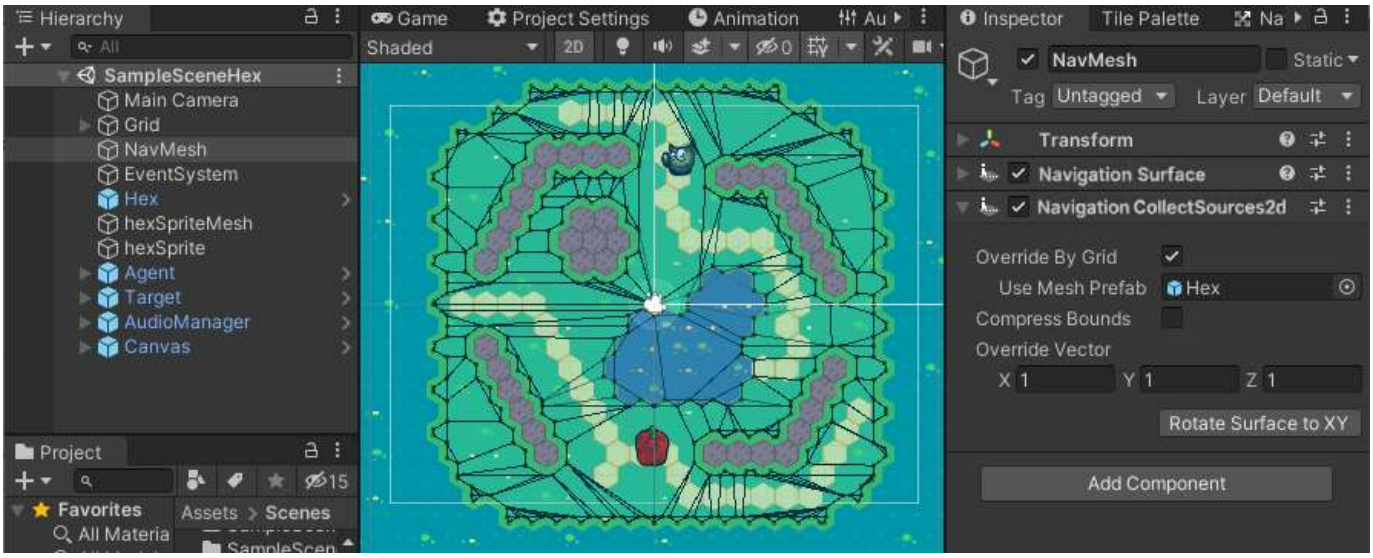


OVERRIDE BY

In some cases the sprite can differ from your grid shape, it may have excessive outline or be concave. When we want clean navigation though the tilemap we need clean to up hundred sprites of different shapes. To streamline tedious editing of each sprites outline we can override sprite mesh altogether.

First we need to have a mesh that has the shape of a tile. For this hexagonal tile example, we'll import a mesh shaped like a hexagon. (Use Blender or any other 3D software)

Add your hexagon to the scene, rotate it to face the camera and scale it down to match the size of your cells. In the `NavigationCollectSources2d` component, check `Override By Grid` and add prefab. Hit Bake.



Important Note: *If we remove the prefab and leave the override option active, all tiles will be considered as squares. Which is great for performance.*

BAKE AT RUNTIME

Often games have procedural world building which requires bake NavMesh at runtime. To achieve this you need to call a single method `BuildNavMeshAsync` in `NavigationSurface` component to bake NavMesh at runtime in Unity async manner. This method can be called on `Start()` method.

```
public NavMeshSurface Surface2D;

void Start()
{
```

```
        Surface2D.BuildNavMeshAsync();
    }
```

For dynamic worlds to save some CPU utilization we can call `UpdateNavMesh` instead of building NavMesh every frame. Code may looks like:

```
    public NavMeshSurface Surface2D;

    void Update()
    {
        Surface2D.UpdateNavMesh(Surface2D.navMeshData);
    }
```

But it does not cache collected 2D sources between updates, so you might use its internal method with your cacheing mechanism:

```
    NavMeshBuilder.UpdateNavMeshDataAsync(data, GetBuildSettings(), sources, sources d
```

Also to save some CPU cycles consider to use `NavigationCollectRootSources2d` to specify navmesh root sources explicitly.

Important note: *Collider's bounds not getting updated to match the transform until 'LateUpdate()' is called. Force the update by calling Physics2D.SyncTransforms() before calling 'BuildNavMesh()' if you baking navmesh at runtime.*

Important note: *Physics updated only in fixed update, that is running in different time period than regular update. So its is not possible to generate physics in Awake/Start and call 'BuildNavMesh()' before 'LateUpdate()' to bake new geometry at runtime.*

More samples can be found in "NavMeshComponents" repository on [GitHub] and `NavMeshBuilder` methods documentation at [Unity] site

KNOWN ISSUES

Prior Unity 6 the agent gets stuck while moving on Y axes, or setting agent. Velocity doesn't do anything if x=0_ - it seems to be a bug in NavMeshAgent component. There is no fix available right now, but there are workarounds. The idea of workaround is simple, adjust Agent movement on X so its never straight up. Navmesh can have a small tilt 89.98 to make agent work. If for some reason its doesnt sove the problem, you need add a drift. Here is a snippet for `SetDestination()` :

```
satic float agentDrift = 0.0001f; // minimal
void SetDestination(GameObject target)
 {
     if(Mathf.abs(transform.position.x - target.transform.position.x) < agentDrift)
     var driftPos = target.transform.position + new Vector3(agentDrift, 0f, 0f);
     agent.SetDestination(driftPos);
 }
```

If you are using Unity 2022.3+, the official "AI Navigation" package changes how the NavMesh backend works. Many users might find that clicking "Bake" results in an empty NavMesh or the blue visualization (Gizmos) doesn't appear at all, even if `FlagAsInSelectionHierarchy` is fixed.

Create a NavMeshAgent FIRST in Unity 2022.3, as the Navigation Surface might fail to bake or visualize properly if there are no defined "Agent Types" or active Agents in the scene. Create a dummy Sprite/GameObject and add a `Nav Mesh Agent` component to it BEFORE you hit the Bake button on your Surface. This ensures the Navigation system initializes the Agent Type settings correctly.

THE END

Try out a fully-featured 2D "run-and-chase" game I made to demo NavMesh 2D: [RedHotSweetPepper].

References:

1. NavMeshSurface - https://docs.unity3d.com/Manual/class-NavMeshSurface.html
2. NavMeshSurface Extensions - https://github.com/h8man/NavMeshPlus
3. The Game Demo - https://github.com/h8man/RedHotSweetPepper
4. Forum to discuss - https://forum.unity.com/threads/2d-navmesh-pathfinding.503596/
5. NavMeshComponents Runtime - https://github.com/Unity-Technologies/NavMeshComponents/tree/master/Assets/Examples/Scenes